STRUCTURAL ERROR — CAPACITY MISALIGNMENT

The system said one thing. The shelf said another.

That was the error. Not the operator. The structure. Here's how we found it — and what it cost us until we did.

WMS CAPACITY VS PHYSICAL REALITY

The system and the shelf disagreed. The shelf won.

During root cause analysis on a cluster of wrong-item picks, we applied the 5 Whys to one recurring location. Every time, the error linked back to a specific shelf that was consistently overcrowded.

The WMS said one thing about shelf capacity. The physical shelf held less.

The system had never been updated to reflect the physical reality. Replenishment was filling to system capacity. The shelf was overflowing. Pick face was a mess. Errors were inevitable.

5 Whys. One location. One structural fix.

1. **Why: wrong item picked**
   Products were stored adjacently with poor visual separation — overcrowded pick face.

2. **Why: pick face was overcrowded**
   Replenishment was filling the location to system-defined capacity.

3. **Why: system capacity exceeded physical capacity**
   WMS capacity figure exceeded what the shelf physically held. The difference was never flagged.

4. **Why: no process to audit system vs physical capacity**
   Capacity was set at system go-live and never reviewed. No trigger for a re-audit.

5. **Root cause: no slotting/capacity review process existed**
   Fix: physical capacity audit → update WMS → standardise replenishment fill limits → schedule recurring reviews.

Capacity Misalignment — System vs Physical Reality

WMS CAPACITY (SYSTEM)

System figure

what the system believed

PHYSICAL CAPACITY (ACTUAL)

Physical reality

what the shelf actually held

RESULT

Systematic overfill

↑ every replenishment cycle

CONSEQUENCES OF SYSTEM–PHYSICAL MISMATCH

Pick face overcrowding → visual confusion → wrong-item picks

Products too close together at the pick face. Operator error that wasn't operator error.

Replenishment inconsistency → empty locations mid-pick

When overfill caused product damage, effective capacity dropped further. Unplanned empty slots.

Fix: physical audit → WMS update → fill limit standardisation → quarterly review

Error rate at affected locations dropped within 2 weeks of correction.

The WMS was set at go-live and never reviewed. That was the real gap.

THE PRINCIPLE

Structural misalignments create errors that look like operator mistakes.

When did you last audit whether your WMS capacity figures match what your shelves actually hold? If it was at go-live — that's worth checking.

Save this if you're chasing errors you can't explain. ↓